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**BACHELOR OF INFORMATION SYSTEMS MANAGEMENT
(CDIM 262)**

**ADVANCED WEB DESIGN DEVELOPMENT AND CONTENT MANAGEMENT
(IMS566)**

GROUP ASSIGNMENT:

HALL BOOKING SYSTEM

PREPARED BY:

No	NAME	NO. MATRIX
1.	NUR IRDIANA BINTI MOKTAR	2024244524
2.	SITI ZULAIKHA BINTI MOHD SALLEH	2024680156
3.	INTAN NURSAKINAH BINTI IDRUS	2024220646
4.	NURUL HIDAYAH BINTI ANUAR	2024680246

CLASS:

CDIM2624ST1

PREPARED FOR:

DR. MUHAMMAD ASYRAF BIN WAHI ANUAR

DATE:

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1.0 Introduction

1.1 Project Background

The booking of halls plays a vital role in planning and arranging various events including meetings, seminars, weddings, conferences, and communal events. The current process of booking the halls is usually done manually through calling, visiting people and through applications. This is not very effective since the customers must wait for a response on the availability of halls while the hall administrators manually manage the information. With the increasing demand for bookings, manual management becomes increasingly hard and may lead to booking clashes, misplacement of information, and scheduling errors. To address all these problems, the Hall Booking System has been created as a web-based software which digitalizes the entire process of hall booking.

1.1.1 Purpose of this system

The purpose of the Hall Booking System is to design an efficient, centralized, and user-friendly interface that can be used to manage the hall reservations. This system is designed to substitute the conventional booking method in which booking can be done through telephone calls, walk in booking, and the messaging application. In such a case, the booking process can face many challenges like booking conflicts, delayed responses, and wrong booking. Through the use of this system, the customers will be in a position to access information about the halls, make reservations, and get confirmation about their bookings in an online manner. On the other hand, the administrators will be able to manage the information of the halls, authenticate booking requests, booking schedule management, receipt of generation, and maintenance of customer information through the database.

1.1.2 Objective

The first goal of the Hall Booking System is to come up with a web-based application that will facilitate and make the booking process easier for the benefit of both customers and administrators. For instance, giving the customer an easy way of seeing the available halls, checking whether the hall can be booked, and making booking requests through the Internet without visiting the booking office physically or contacting the administrator. Additionally, the system also gives customers the chance of receiving booking confirmation and managing booking information easily.

The other objective of the system is to provide the administrator with convenience of managing hall information, managing booking schedules, verifying booking requests, and even issuing booking receipts. This will help in minimizing paperwork and booking conflicts as well. In summary, the Hall Booking System aims to provide a fast and efficient booking process.

1.2 System Analysis

The Hall Booking System has been created to solve the drawbacks associated with the conventional booking of halls, which involves the development of a centralized web-based booking management system. The current methods used in the booking of hall include manual booking, which involve using phones, visiting the offices in person, completing forms or using messenger applications, and all these methods result in conflicting bookings, late responses and inability to track the booking history.

Customers will be able to register and make bookings while administrators will be able to manage hall details and approve customers' bookings using this system. Through computerization of the booking process, both administrators and customers will have an easier time managing their booking affairs.

1.2.1 Problem statement

A lot of halls of booking systems operate manually, which leads to many issues on the part of both the customer and the administrators. The customers will have to reach out to the management using phone calls, messages via WhatsApp or even physically visiting the hall to inquire whether the hall is available for booking. This is time-consuming and the customers will never get an idea about available booking dates in real time.

As far as the administrators are concerned, manual booking increases the chances of human error such as double bookings, erroneous customer details, and missing booking records among others. It becomes harder to manage the payment status, updating booking schedule, and generation of receipts where booking is made manually.

The hall booking system helps solve these problems since it allows the creation of an online booking system. The customer will be able to know about the availability of the halls, booking, and even keep track of the booking status from within the system. At the same time, the administrator will be able to manage all details, confirm bookings, manage bookings, and even make receipts.

1.2.2 Intended Users

This Hall Booking System is intended for two types of users which are customers and administrators. Firstly, the intended users of this system are customers. Customers can create accounts, login into the system, explore halls available for booking, see information about the halls such as the number of people the hall can accommodate and cost of hiring the hall, choose a particular day and time for booking and finally book the hall. They can also check their booking history through this system.

Secondly, the other intended users of this system are the administrators. Administrators are those who oversee the general functioning of the system. They can manage customer accounts, add or edit the information of the halls, verify bookings, approve or reject bookings, update booking schedules, check booking records, and produce booking receipts.

1.3 Github Repository Link

<https://github.com/Nurirdiana/Hall-Booking-System>

2.0 User Role / Description

2.1 Admin

The admin is responsible for managing the overall system of operations.

Functions of admin include:

- Managing hall information and availability
- Handling customer bookings
- Verifying booking
- Updating booking schedules
- Monitoring booking

2.2 Customer

The customer uses the system to make hall reservations.

Functions of customer include:

- Registering and logging into the system
- Viewing hall details and availability
- Making bookings
- Receiving booking updates

3.0 Main Functions

To support its two main user roles, Admin and Customer. The Hall Booking System was designed with a clear distribution of functions. This division guarantees the system's efficiency, security, and usability. The primary functions are grouped according to the role of the user and how they interact with the core elements of the system.

3.1 Systems Requirements

3.1.1 Software Requirements

The Hall Booking System's minimal hardware and software requirements.

For Admin & Customer (Web Browser):

Requirement	Minimum Specification
Operating of system	Windows 10/11, MacOS, Linux, Android, iOS
Internet	Minimum 1 Mbps
Web Browser	Latest Google Chrome, Mozilla Firefox, Microsoft Edge, Safari
Screen Resolution	1024 * 768 or higher
RAM	Recommend 4GB

Table 1: Requirement for Web Browser

For Server (Hosting):

Requirement	Minimum Specification
Web Server	Apache 2.4
PHP	PHP 7.4 or higher
Database	MySQL 5.7/higher, MariaDB 10.4+
Server RAM	2GB minimum
Storage	1 GB free space minimum

Table 2: Requirement for Hosting

3.1.2 Version Compatibility

To make sure every part of the system functions properly

Browser Compatibility

Browser	Supported Versions	Status
Google Chrome	Version 90 +	Fully Supported

Microsoft Edge	Version 88 +	Fully Supported
Mozilla Firefox	Version 90 +	Fully Supported
Safari	Version 14 +	Supported

Table 3: Browser Compatibility

Server Compatibility

Component	Complete Version
PHP	7.4, 8.0, 8.1, 8.2
MySQL	5.7, 8.0
MariaDB	10.4, 10.5, 10.6

Table 4: Server Compatibility

For best performance and security, use the latest stable versions

3.1.3 Dependencies

external frameworks, libraries, or tools required to run the system.

Backend Dependencies:

Dependency	Purpose
PHP	Backend logic for core programming language
MySQL/MariaDB	To store user, hall, booking data database

Table 5: Backend Dependencies

Frontend Dependencies:

Dependency	Purpose
HTML5	Web pages structure
CSS3	Layout and styling
JavaScript	Interactive features (forms, validation)
Bootstrap 5	Responsive design framework

Table 6: Frontend Dependencies

Security:

Dependency	Purpose
bcrypt	Password encryption

Table 7: Security Dependencies

3.2 Features and Functionalities

3.2.1 User Authentication

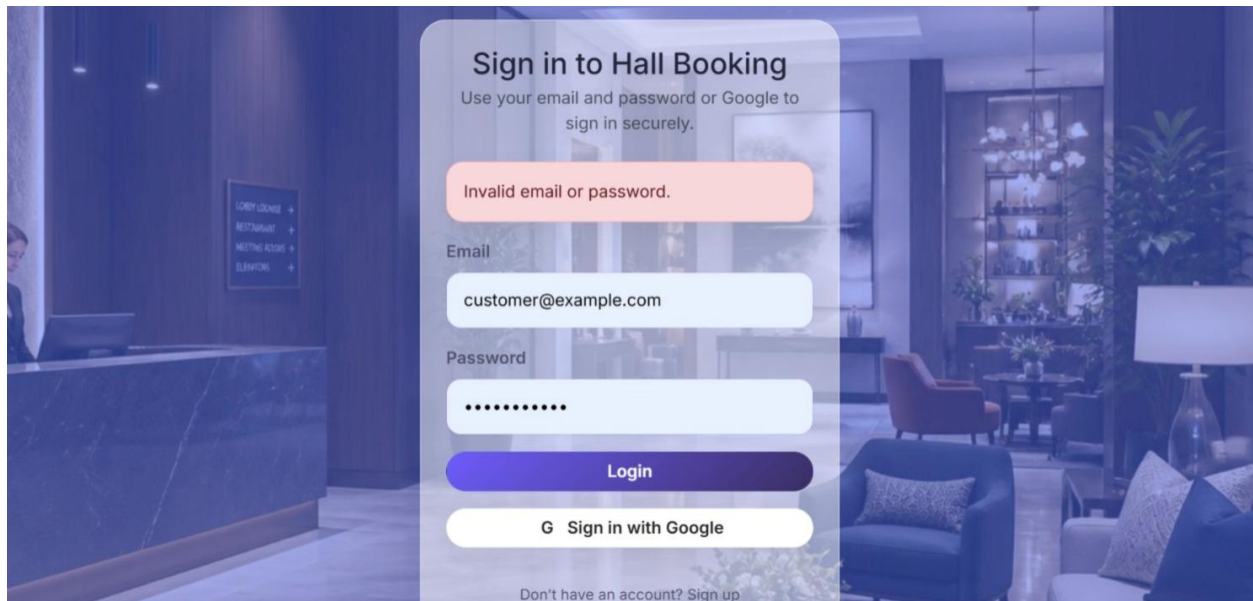


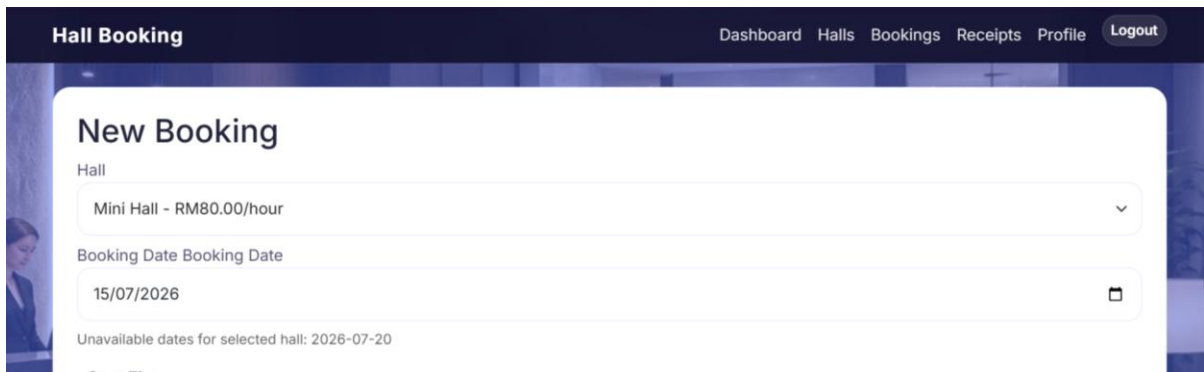
Figure 1: Invalid email or password at login page

User authentication is the entrance to the Hall Booking System. It guarantees:

- **Security:** Only authorized users can access the system
- **Personalization:** Each user sees specific statistics and dashboard
- **Role Management:** Different users (Admin vs Customer) have different authorizations
- **Account Protection:** Passwords are encrypted and login attempts are validated

Users must validate their identity (via email + password) before they can access any section of the system, assuring data security and personalized experience

3.2.2 Real Time Booking Validation



The screenshot displays a web interface for 'Hall Booking'. At the top, there is a navigation bar with links to 'Dashboard', 'Halls', 'Bookings', 'Receipts', 'Profile', and a 'Logout' button. The main content area is titled 'New Booking'. It contains two input fields: a dropdown menu for 'Hall' showing 'Mini Hall - RM80.00/hour' and a date picker for 'Booking Date' showing '15/07/2026'. Below the date picker, a message indicates 'Unavailable dates for selected hall: 2026-07-20'.

Figure 2: Date validation

When a customer chooses a date and time, the system immediately verifies and displays if a hall is available or previously reserved. This is known as real-time booking validation. By verifying availability in real time, the system can avoid duplicate bookings.

Important Features of Validation

- Hall Selection: The customer must first choose a hall. The Date Checking System looks for current reservations in the booking table.
- Dates with booked reservations are indicated as unavailable.
- Time checking keeps starting and end times from overlapping.
- Instant Feedback: Users can view validation findings right away.

3.2.3 Create Booking



The screenshot shows a web form titled "New Booking". It contains several input fields: a dropdown menu for "Hall" with the placeholder "-- Select Hall --"; a date input field for "Booking Date" with the placeholder "dd/mm/yyyy" and a calendar icon; a text input field for "<Start Time" with the placeholder "--:-- --" and a clock icon; an "End Time" field with the same placeholder and clock icon; and a "Purpose" dropdown menu with options "PENDING", "CONFIRMED", and "CANCELLED", where "PENDING" is currently selected. Below the form, there is a small text note: "Select a hall to see unavailable dates."

Figure 3: Create Booking for Admin

Create Booking for Admin

Purpose: The admin can make reservations for customers (particularly helpful for walk-in customers or phone reservations).

Features:

- The admin chooses a customer from the database.
- The admin selects the time, date, and hall.
- The booking status can be easily set by admin (Pending, Confirmed, Cancelled).
- All booking details are fully under the admin's control.
- By simply confirming reservations, admin can avoid the approval process.

The screenshot shows a 'New Booking' form with the following elements:

- Hall:** A dropdown menu with the placeholder text '-- Select Hall --'.
- Booking Date:** A text input field with the placeholder 'dd/mm/yyyy' and a calendar icon on the right.
- Start Time:** A text input field with the placeholder '<Start Time' and the format '--:-- --'.
- End Time:** A text input field with the format '--:-- --'.
- Purpose:** A large text area for entering the reason for the reservation.
- Buttons:** A blue 'Save' button and a light blue 'Back' button at the bottom left.
- Feedback:** A small error message 'Please fill out this field.' is visible next to the Hall dropdown.

Figure 4: Create Booking for Customer

Create Booking for Customer

Purpose: Customers are able to reserve halls for themselves.

Features:

- From the available alternatives, the customer chooses a hall.
- The customer can choose the time, date, and enter the reason for the reservation.
- The booking is submitted by the customer for approval by the admin.
- The booking status is automatically set to PENDING and cannot be changed by the customer.
- Customers are only able to view and control their own reservations.

3.2.4 Double-Booking Prevention

Hall Booking Dashboard Halls Bookings Receipts Profile Logout

New Booking

Hall
Mini Hall - RM80.00/hour

Booking Date Booking Date
20/07/2026

Unavailable dates for selected hall: 2026-07-20 The selected date is unavailable.

<Start Time
10:00 AM

End Time
05:00 PM

Purpose

Figure 5: Double Booking Prevention for the same date

The date will automatically become unavailable if there is a customer that books the same hall for the same date as a previous booking to prevent double reservations.

3.2.5 View Booking

Search Reset

Customer	Hall	Date	Time	Total	Status	Action
Demo Customer	Mini Hall	2026-07-20	10:00:00 - 17:00:00	RM 560.00	CONFIRMED	View

Figure 6: View button to view booking details

To view more details about the booking, user can click view button in booking list

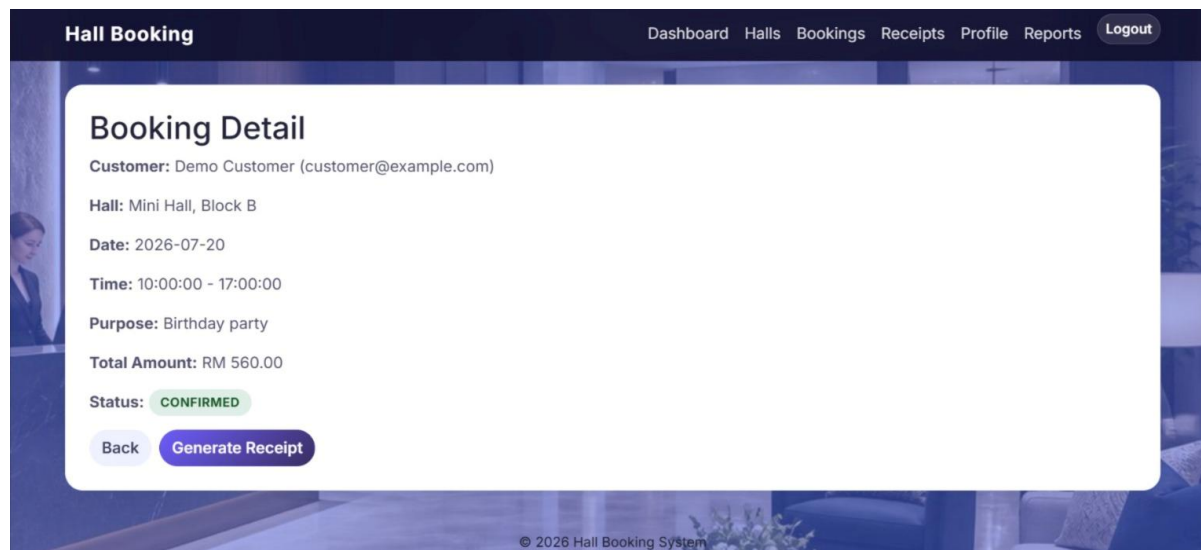


Figure 7: Booking Details for Admin

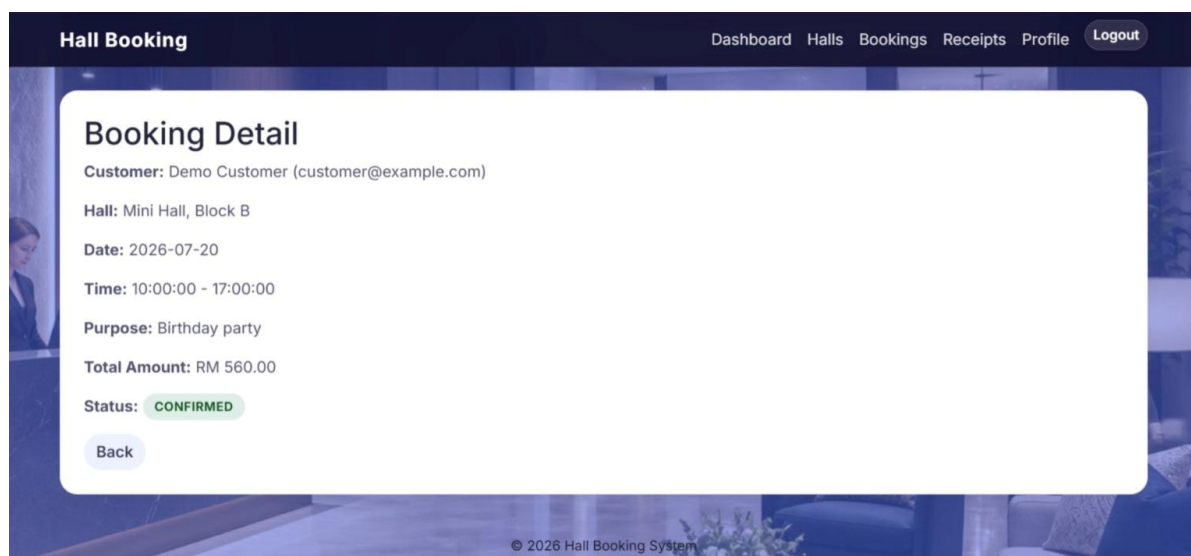
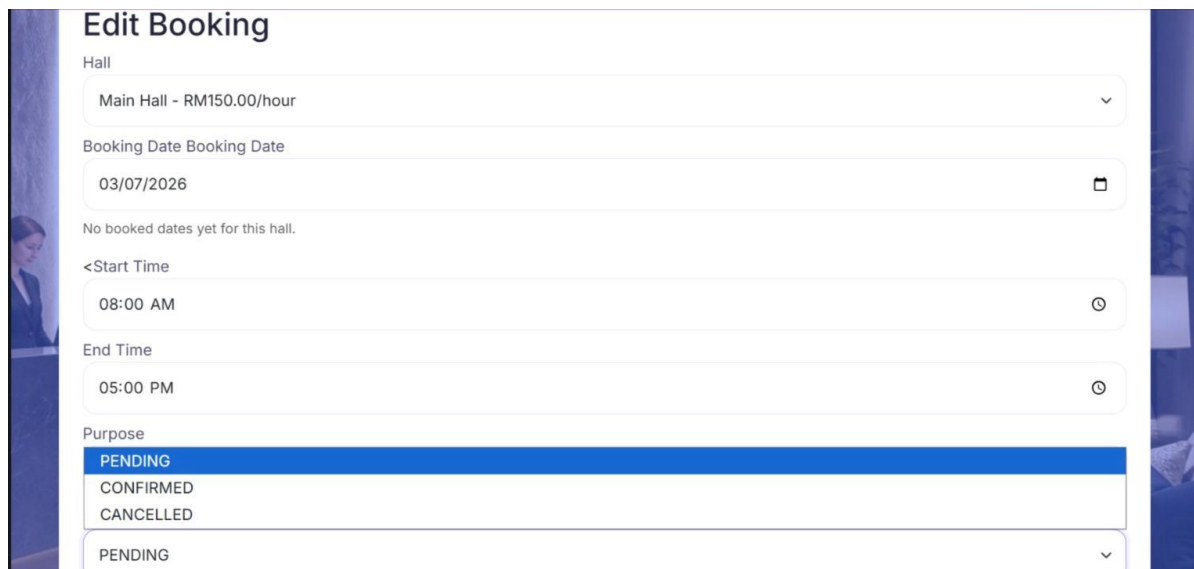


Figure 8: Booking Details for Customer

3.2.6 Update Booking



The 'Edit Booking' form contains the following fields and options:

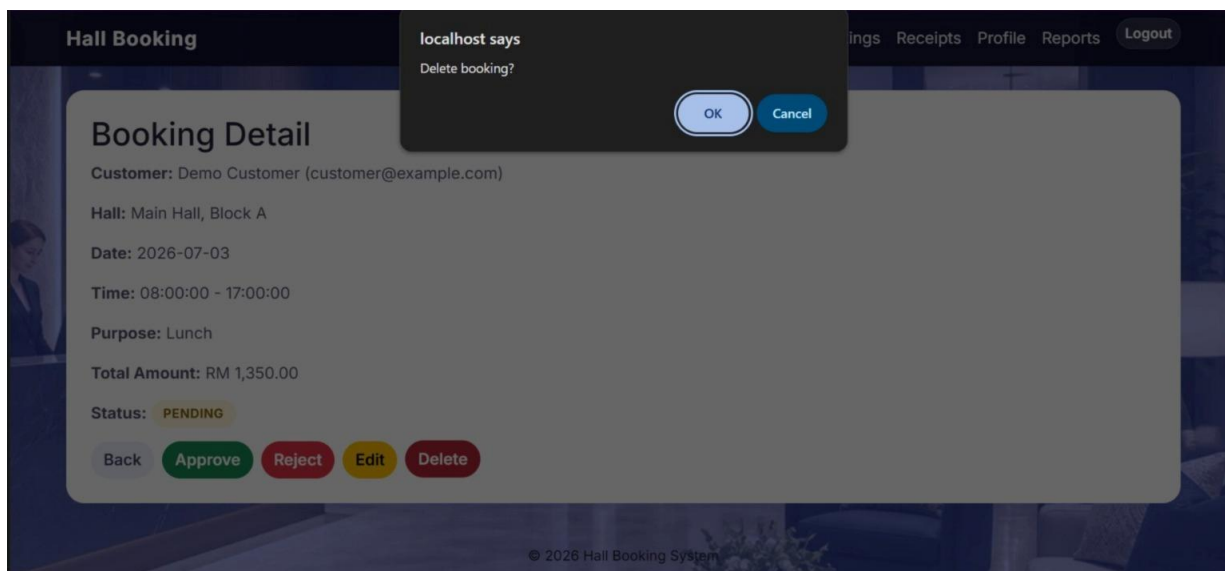
- Hall:** A dropdown menu showing 'Main Hall - RM150.00/hour'.
- Booking Date:** A date input field showing '03/07/2026' with a calendar icon.
- Start Time:** A time input field showing '08:00 AM' with a clock icon.
- End Time:** A time input field showing '05:00 PM' with a clock icon.
- Purpose:** A dropdown menu with options: 'PENDING' (selected), 'CONFIRMED', and 'CANCELLED'.

Below the form, there is a message: 'No booked dates yet for this hall.'

Figure 9: Edit Booking Features for Admin

When necessary, admin may update the current booking details. This is helpful for fixing mistakes or fulfilling any customer requirements. Bookings can only be edited by the admin. Customers are unable to change their reservations as they can only view them. Customers must get in touch with the admin if they need to modify their reservation.

3.2.7 Delete Booking



The screenshot shows a 'Delete booking?' confirmation dialog box with 'OK' and 'Cancel' buttons. In the background, the 'Booking Detail' form is visible, displaying the following information:

- Customer:** Demo Customer (customer@example.com)
- Hall:** Main Hall, Block A
- Date:** 2026-07-03
- Time:** 08:00:00 - 17:00:00
- Purpose:** Lunch
- Total Amount:** RM 1,350.00
- Status:** PENDING

At the bottom of the booking detail form, there are five buttons: 'Back', 'Approve', 'Reject', 'Edit', and 'Delete'.

Figure 10: Delete booking function for Admin

Booking records can be permanently deleted from the system by admin. This can be used to clean up test data, eliminate duplicate entries, or remove accidentally created bookings. Delete is permanent and if admin wants to maintain the record but label it as cancelled, they should always use reject rather than delete function.

Difference: Delete vs Reject function

Reject: Booking status will change to “Cancelled”

Delete: Booking will permanently remove from the system records

3.2.8 Implied Functionalities

- Role-based access control

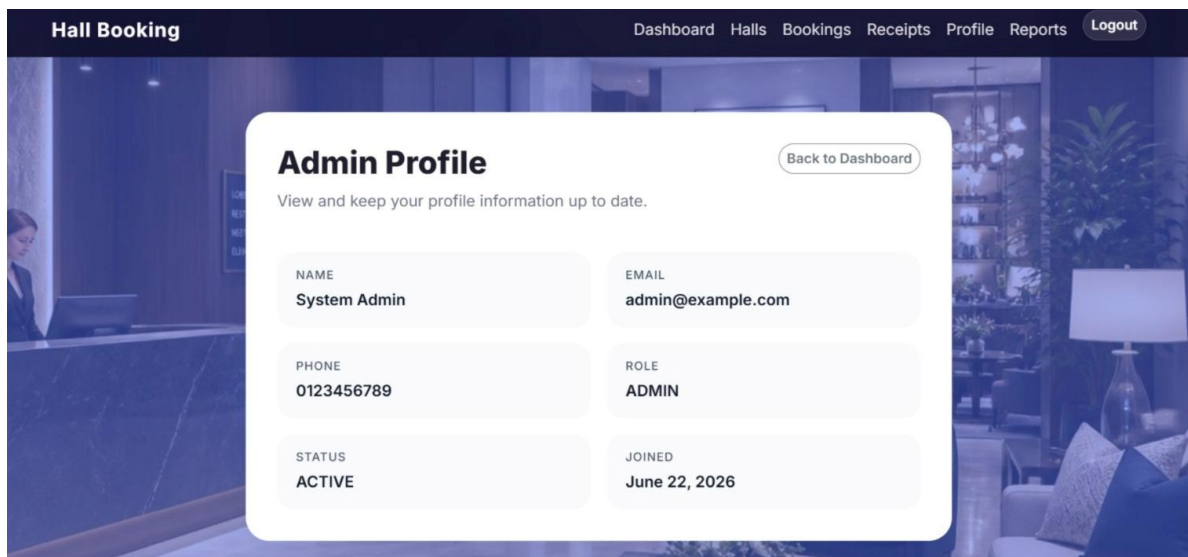


Figure 11: Admin Profile

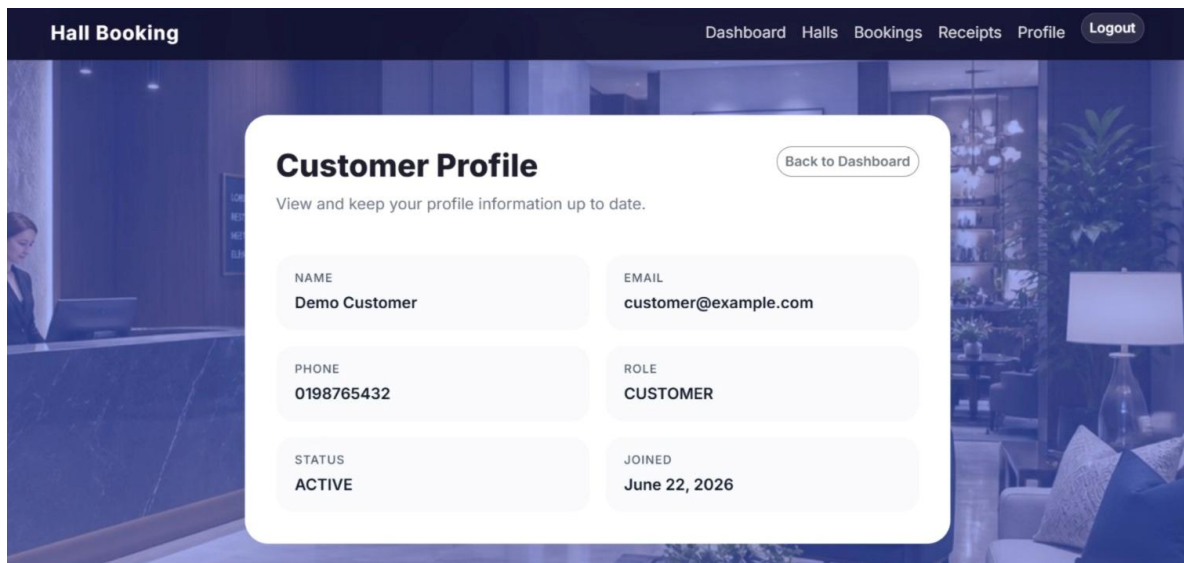


Figure 12: Customer Profile

- Booking Status

Booking List + New Booking

All Search booking dd/mm/yyyy All Status

Search Reset

Customer	Hall	Date	Time	Total	Status	Action
Demo Customer	Mini Hall	2026-07-20	10:00:00 - 17:00:00	RM 560.00	CONFIRMED	View
Demo Customer	Main Hall	2026-07-03	08:00:00 - 17:00:00	RM 1,350.00	PENDING	View
Demo Customer	Mini Hall	2026-06-22	08:00:00 - 12:00:00	RM 320.00	CANCELLED	View

Figure 13: Booking Status

- PDF Receipt export

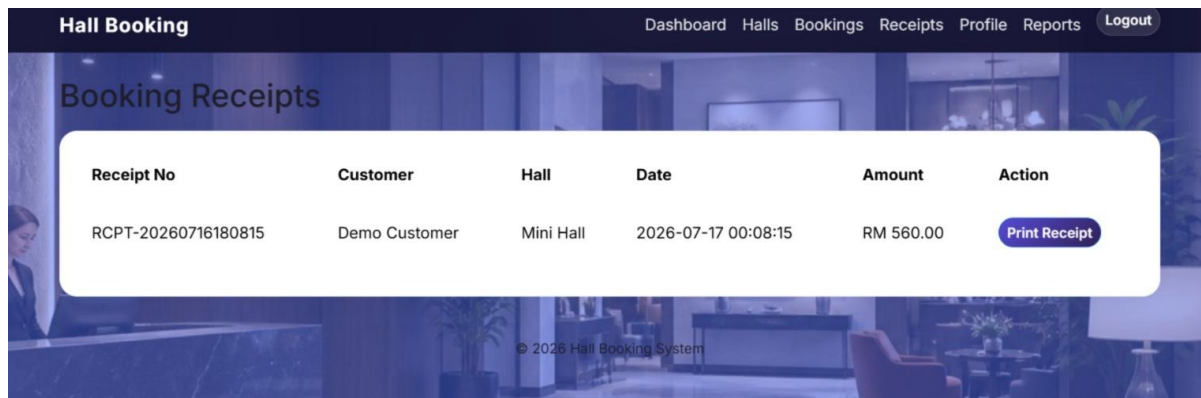
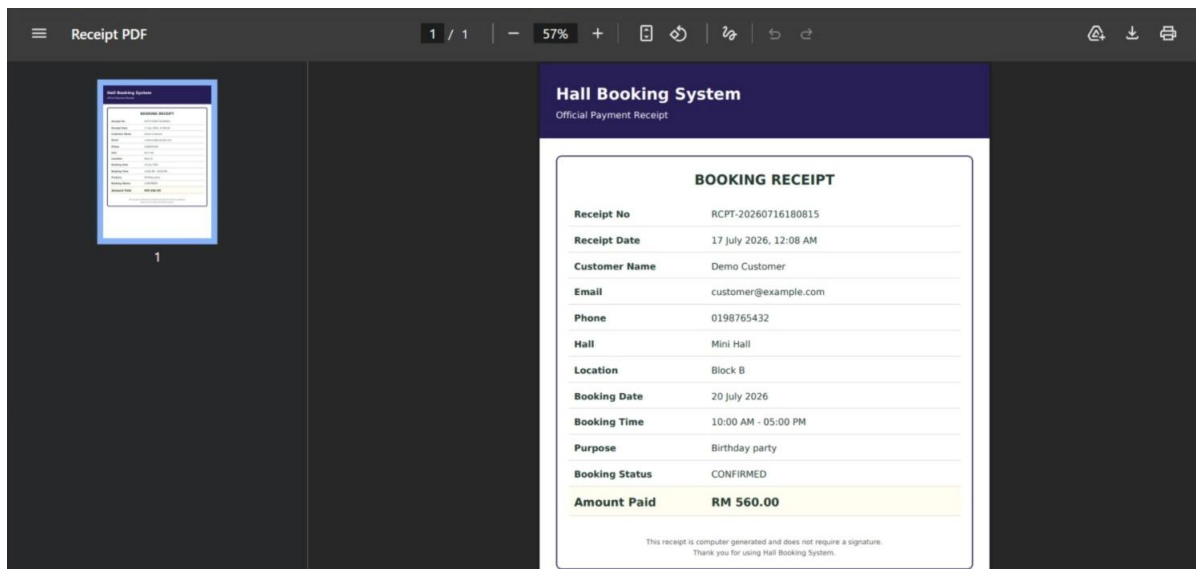


Figure 14: Print Receipt to PDF function



3.3 Use Case Diagrams / User Stories

Use case diagrams is essential for illustrating how users interact with the system. They guarantee that all user requirements are recorded and help in defining the project's scope. The detailed user stories for admin and customer navigation are listed here.

3.3.1 Admin Navigation and Usage

Admin can create, read, edit, and manage hall booking information.

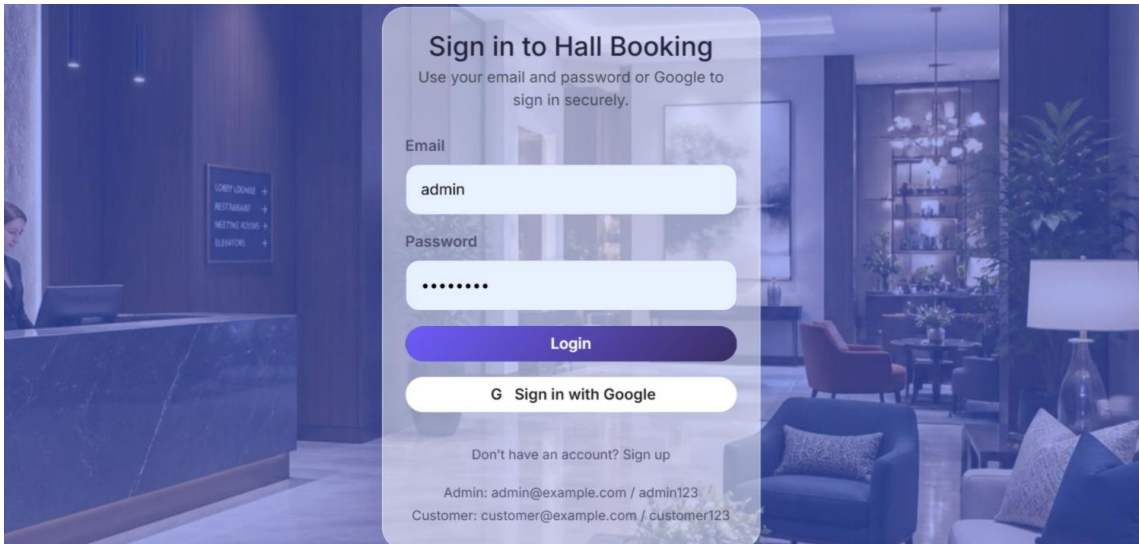
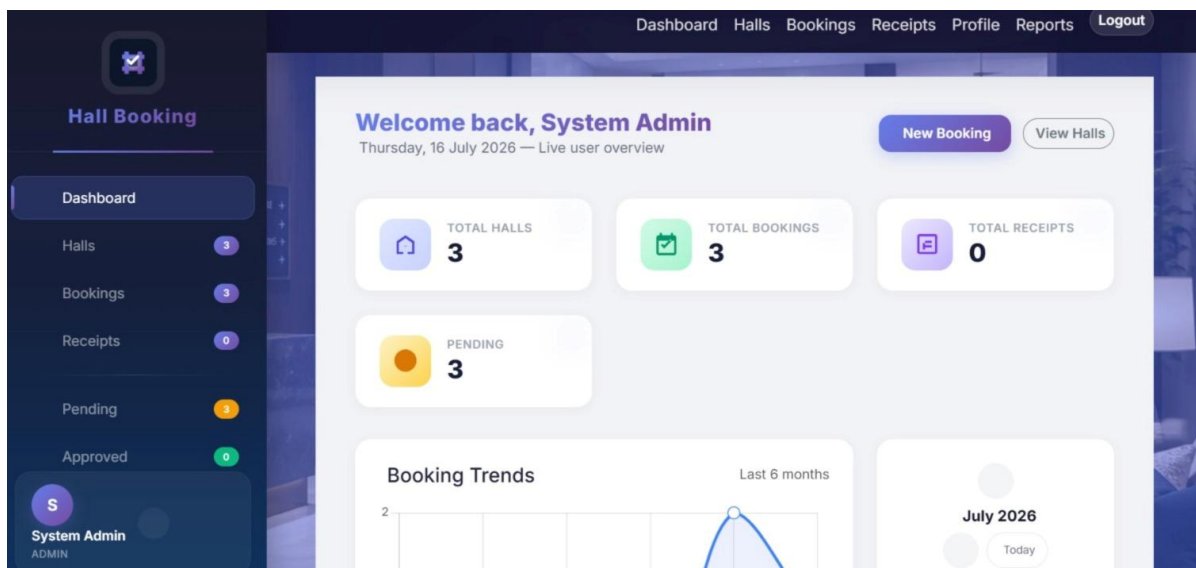


Figure 15: Admin Login Page

Admin Login Page Flow:

1. **Action:** Admin goes to the login page of the system.
2. **System Response:** A secure login form with email and password fields, and a "Login" button is displayed by the system.
3. **Action:** Admin inputs their login information, such as admin@example.com or admin123.
4. **System Response:** The system verifies the admin role and redirects it to the "Admin Dashboard" after successful authentication.



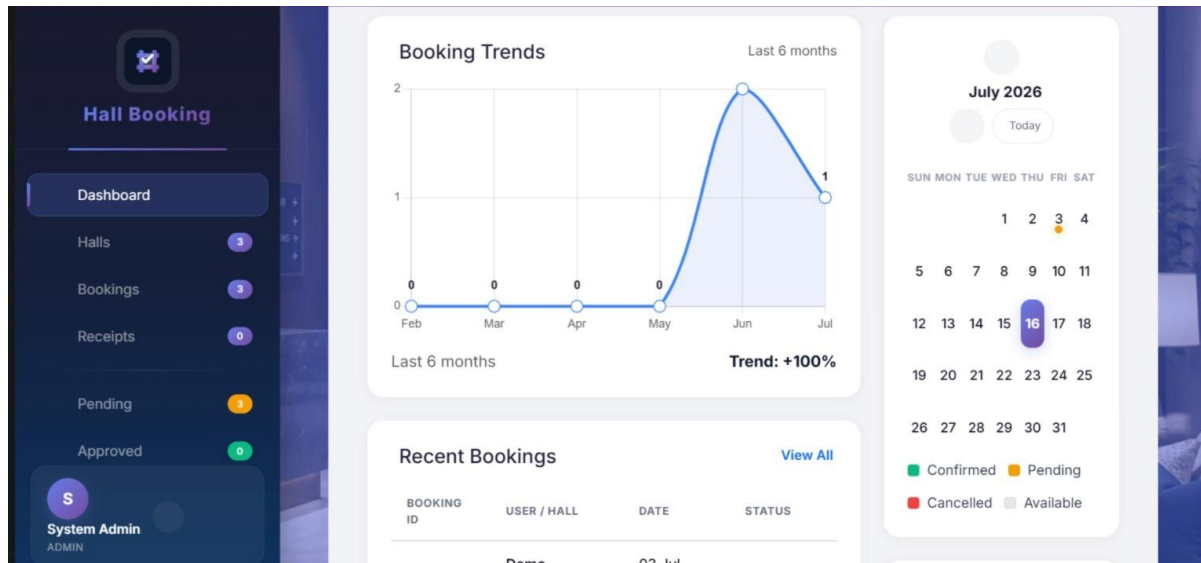


Figure 16: Admin Dashboard

Admin Dashboard Flow:

1. **Action:** Admin is taken to the main dashboard after successfully logging in.
2. **System Response:**
 - The dashboard offers a high-level summary of recent activity and system information such as the admin name, and the current date is displayed in the welcome message.
 - The left sidebar has a quick navigation menu that includes Dashboard, Halls, Bookings, Receipts, Pending, and Approved while 3 total reservations are displayed on the Total reservations counter.
 - The total receipts counter display four receipts. A calendar view with booking status indicators for July 2026 and the most recent booking activity are displayed in the "Recent Bookings" table.
3. **Action:** To access particular management sections, admin can click on any menu item.
4. **System Response:** Every click takes users to the relevant administration page.

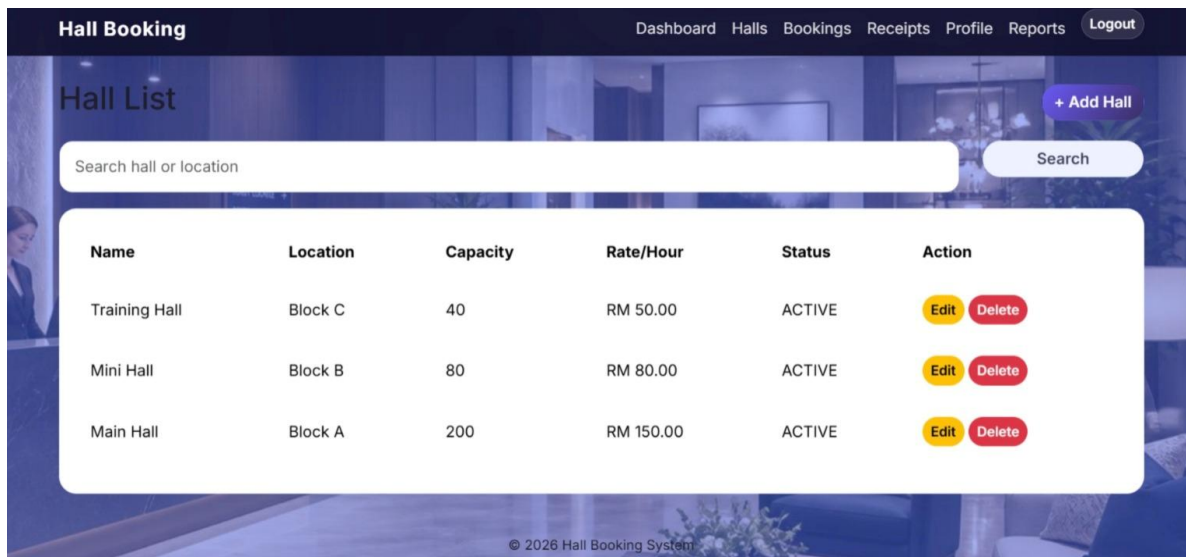


Figure 17: Hall List

Hall List Management Flow:

1. **Action:** Admin selects "Halls" from the sidebar menu.
2. **System Response:** All registered halls and their information are displayed on the Hall List page.
3. **Action:** Admin can locate halls by name or location using the search bar.
4. **Action:** To create a new entry, click the "+ Add Hall" button.
5. **Action:** Click "Edit" to modify the hall's capacity, rate, description, and status.
6. **Action:** To delete a hall from the system, click "Delete" (a confirmation prompt will appear).
7. **System Response:** Search for real-time filtering and display of matching halls. Add Hall: This form allows users to enter new hall information.
Edit: This opens a pre-filled form with the hall's current information.
Delete: After admin approval, the hall record is deleted

Customer	Hall	Date	Time	Total	Status	Action
Demo Customer	Main Hall	2026-07-03	08:00:00 - 17:00:00	RM 1,350.00	PENDING	View
Demo Customer	Mini Hall	2026-06-22	08:00:00 - 12:00:00	RM 320.00	PENDING	View
Demo Customer	Mini Hall	2026-06-22	08:00:00 - 12:00:00	RM 320.00	PENDING	View

Figure 18: Booking List

Booking List Management Flow:

1. **Action:** Admin will select "Bookings" from the sidebar menu.
2. **System Response:** All booking records are shown on the Booking List page.
3. **Action:** Bookings can be filtered by admin using Date picker (format: mm/dd/yyyy), filter by status (All, Confirmed, Pending, Cancelled) and to locate specific reservations, use the search bar.
4. **Action:** To manually make a reservation (for walk-in clients), click the ". New Booking" button.
5. **Action:** Click "View" to update the status and view complete booking details.
6. **Action:** To create a report of all reservations, click "Export Booking PDF".
7. **Action:** Click "Reset" to display all bookings and remove all filters.
8. **System Response:**
 - Filters: Modifies the table according to predetermined standards
 - New booking: the booking creation form will be open.
 - View: Shows all booking information along with options for approval or rejection
 - Export: generates a PDF report that can be downloaded.

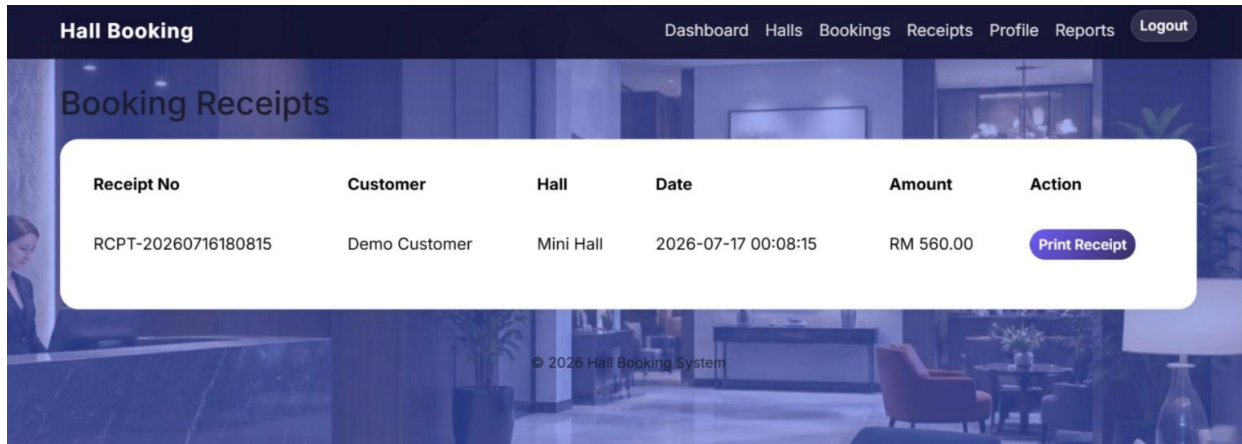


Figure 19: Booking Receipts

Booking Receipts Management Flow:

1. **Action:** Admin will select "Receipts" from the sidebar menu.
2. **System Response:** The Booking Receipts page is shown by the system.
3. **Action:** To create a printable receipt for client records, admin can click "Print Receipt".
4. **System Response:** All booking details, payment information, and system validation are included in the formatted receipt that the system produces.

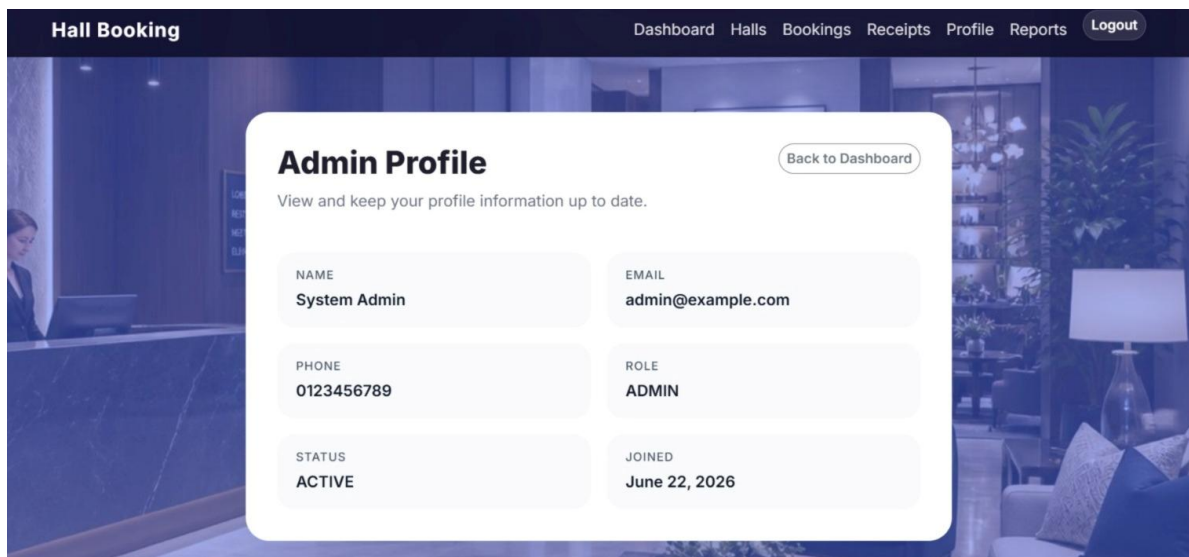


Figure 20: Admin Profile

Admin Profile Management Flow:

1. **Action:** Admin selects "Profile" from the sidebar menu.

2. **System Response:** All account details are displayed on the admin profile page.
3. **Action:** Admin has access to change profile details (Name, Email, Phone).
4. **Action:** Click "New Booking" to make a reservation straight from user profile.
5. **Action:** Click "My Bookings" to see user's own booking history.
6. **System Response:**
 - Updates with admin_id are stored in the user's table.
 - Redirects to the relevant booking creation or view pages

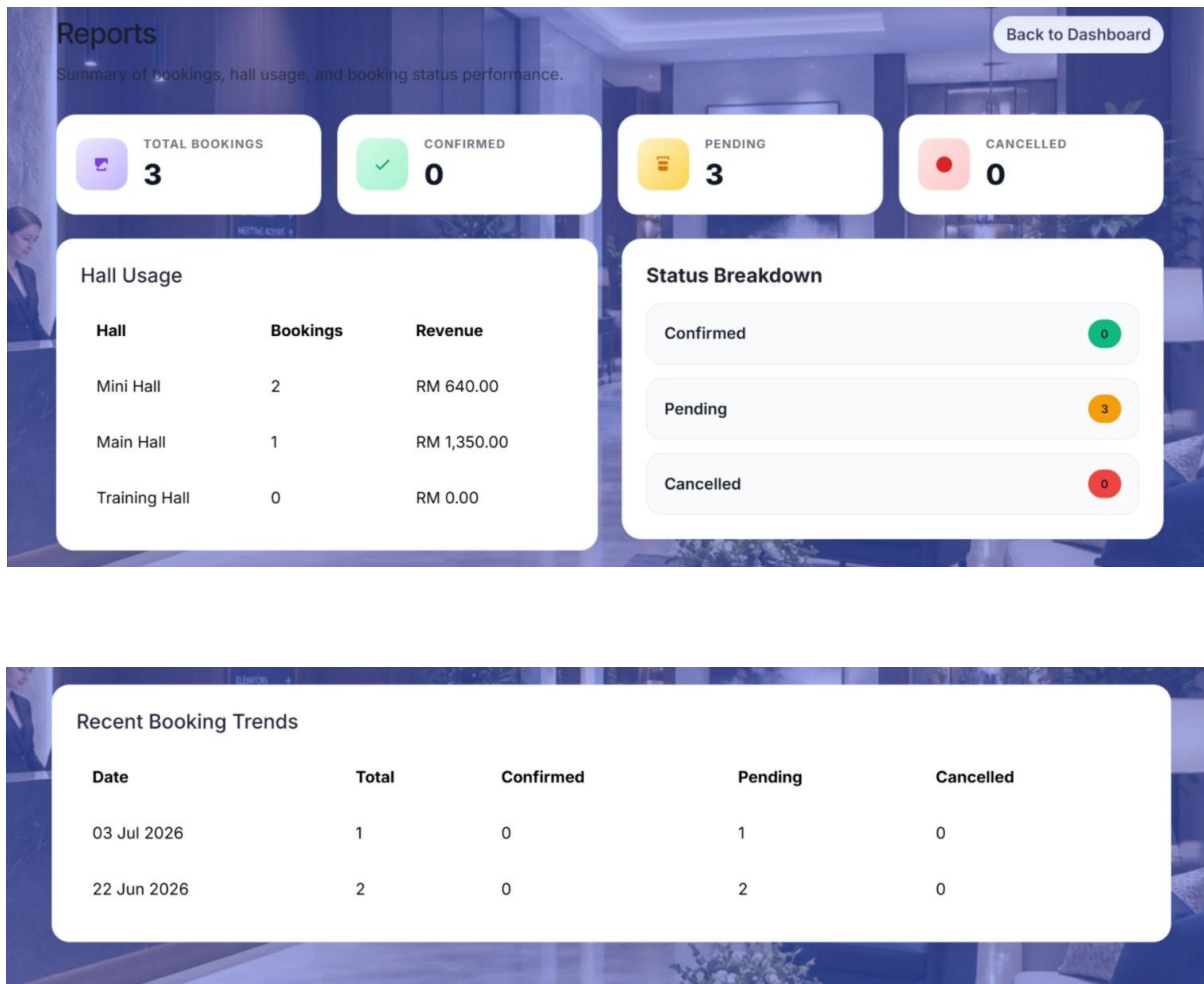


Figure 21: Reports and Analytics

Reports and Analytics Flow:

1. **Action:** Admin selects "Reports" from the sidebar menu.

2. **System Response:** The analytics dashboard and detailed information are shown by the system.

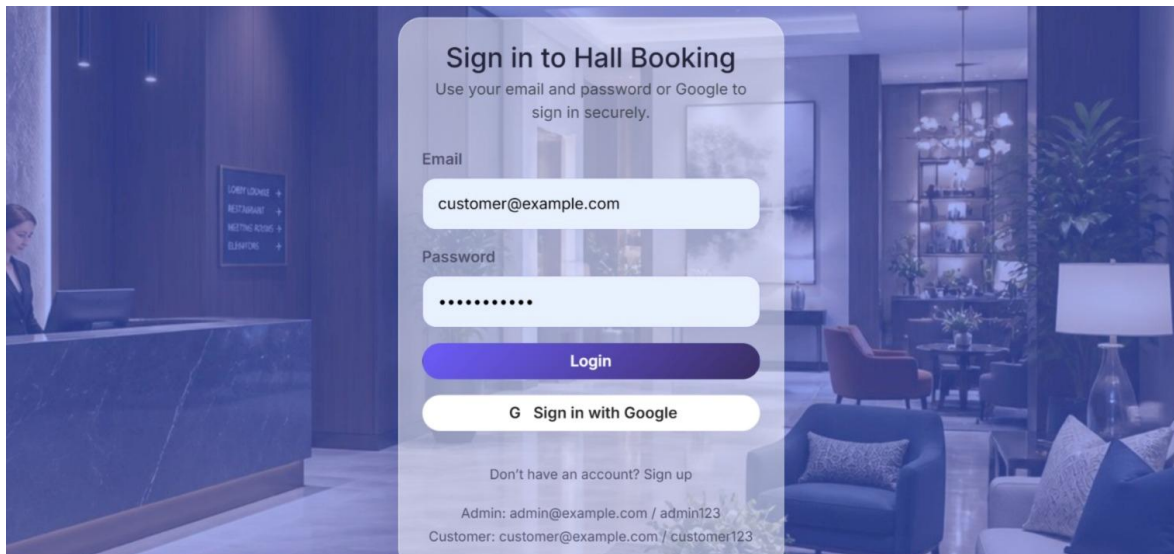
Metric	Value
Total Bookings	3
Confirmed	0
Pending	3
Cancelled	0

Table 1: Key Metrics Displayed

3. **Action:** Admin can see recent booking trends and status breakdowns.
4. **System Response:** The bookings, halls, and booking_receipts columns are used to automatically compute all data.
5. **System Response:** Visualizations and charts show:
- Status breakdown (percentages of confirmed, pending, and cancelled)
 - Hall utilization rates
 - Recent booking trends over time.

3.3.2 Customer Navigation and Usage

Customers can select venues, choose booking dates and times, edit, and submit booking requests.



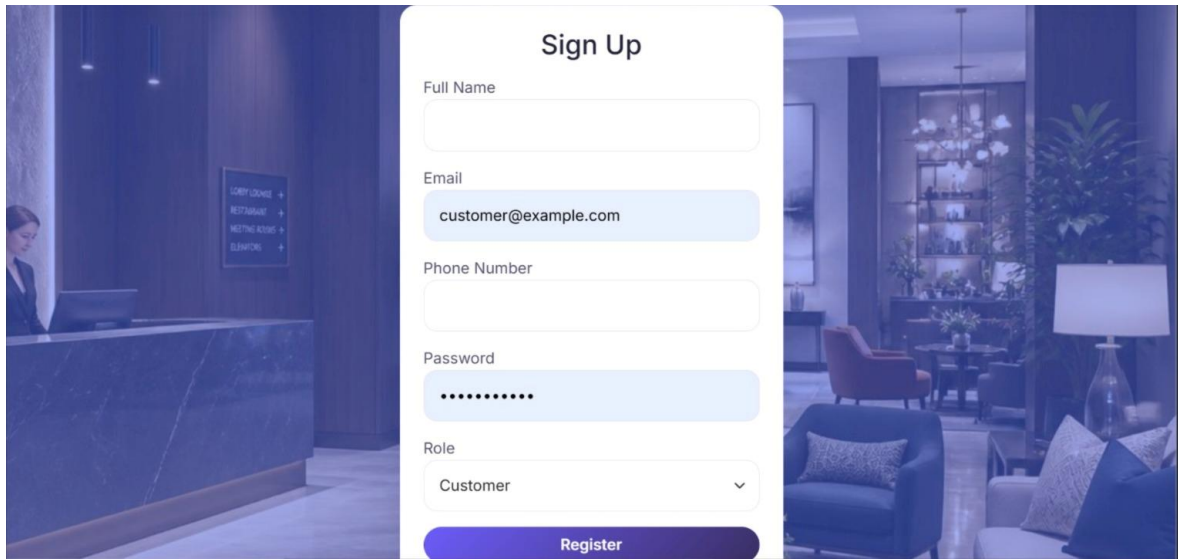


Figure 22: Customer Sign Up and Login Page

Customer Sign Up Flow:

1. **Action:** A new user accesses the system and selects the "Sign Up" option on the login screen.
2. **System Response:** The registration form with the following fields is shown by the system: Complete Name, Email, Phone Number, Password, and Role (which is set to "Customer")
3. **Action:** The customer enters the necessary data.
4. **Action:** The customer presses the "Register" button.
5. **System Response:**
The system verifies that all fields are filled out accurately, determines whether the email address is already registered in the users table, creates a new user record with role set to "Customer" and approval_status set to "Active," directs the customer to the login page with a success message, and allows them to log in with their credentials.
6. **Note:** Additionally, customers may select "Already have an account? To go back to the login screen, click "Login."

Customer Login Flow:

1. **Action:** The customer navigates to the login page of the system.
2. **System Response:** A secure login form with email and password fields is displayed by the system.
3. **Action:** Customer inputs their login information, such as customer@example.com or customer123.

4. **System Response:** After authentication is successful, the system verifies that the user's role is "Customer," with their approval status is "Active", their credentials are valid against the users table, and lastly, they are redirected to the Customer Dashboard.

Demo Credentials:

Customer: customer123 or customer@example.com

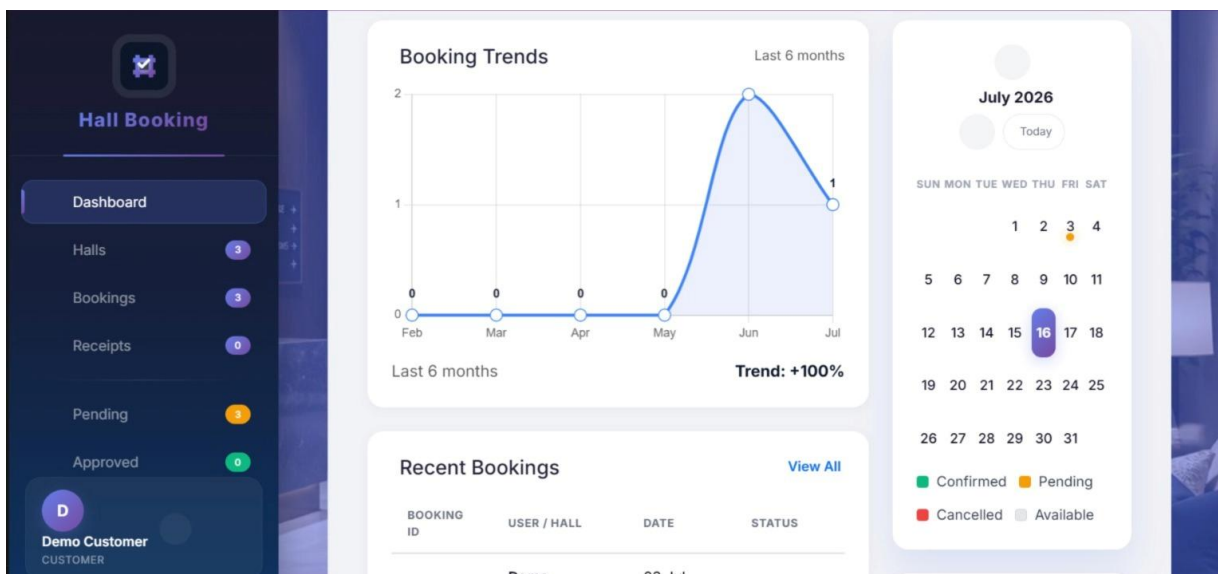
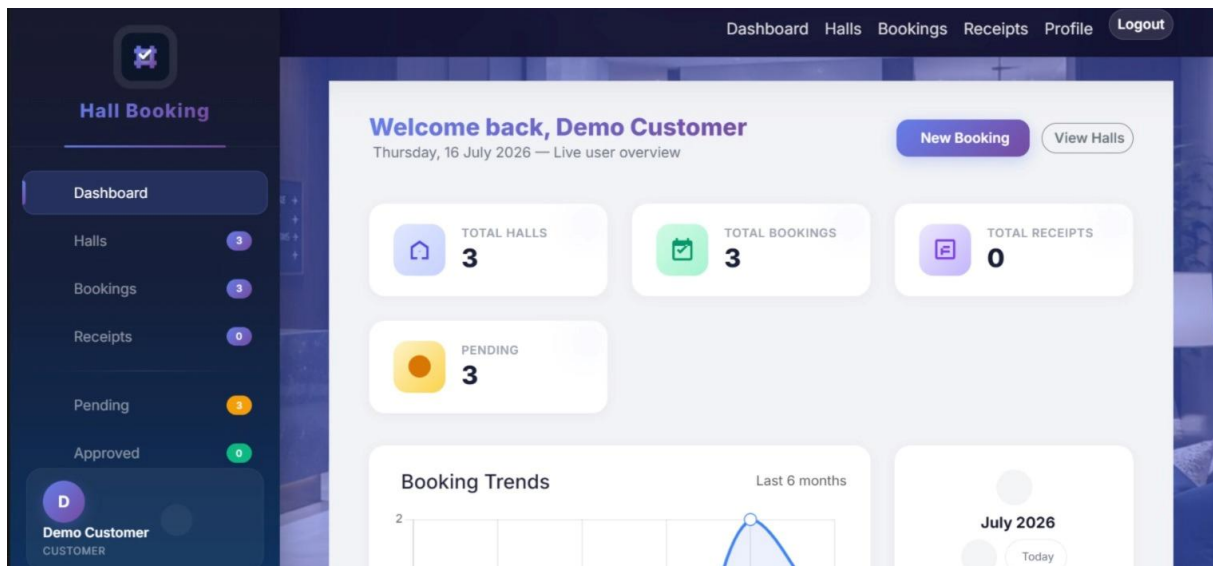


Figure 23: Customer Dashboard

Customer Dashboard Flow:

1. **Action:** The customer is taken to the main dashboard after successfully logging in.
2. **System Response:** The dashboard offers a customized summary that includes:
 - The customer's name and current date are displayed in the welcome message. The quick navigation menu (Dashboard, Halls, Bookings, Receipts, Profile, Logout) is located on the left sidebar.
 - The total metrics displayed are: Total Bookings (3), Pending Bookings (2), Total Receipts (1), Recent Bookings table (which displays the most recent booking actions)
 - Calendar view (which shows July 2026 with booking indicators)

Name	Location	Capacity	Rate/Hour	Status	Action
Training Hall	Block C	40	RM 50.00	ACTIVE	Book
Mini Hall	Block B	80	RM 80.00	ACTIVE	Book
Main Hall	Block A	200	RM 150.00	ACTIVE	Book

Figure 24: Hall List (Customer View)

Hall List Management for customer flow:

1. **Action:** Customer clicks on "Halls" from the sidebar menu.
2. **System Response:** The system will display the Hall List page showing all available halls with their details.
3. **Action:** Customer may use the search bar to find specific halls by name or location.
4. **Action:** Click the "Book" button on any hall to proceed with booking.
5. **System Response:**
 - Search: Real-time filtering and display of matching halls
 - Booking: Opens the booking creation form for the selected hall

Customer	Hall	Date	Time	Total	Status	Action
Demo Customer	Main Hall	2026-07-03	08:00:00 - 17:00:00	RM 1,350.00	PENDING	View
Demo Customer	Mini Hall	2026-06-22	08:00:00 - 12:00:00	RM 320.00	PENDING	View
Demo Customer	Mini Hall	2026-06-22	08:00:00 - 12:00:00	RM 320.00	PENDING	View

Figure 25: Booking List (Customer View)

Booking List Management Flow:

1. **Action:** Customer clicks on "Bookings" from the sidebar menu.
2. **System Response:** The system displays the customer's personal Booking List page.
3. **Action:** Customer can filter bookings using:
 - Date picker (mm/dd/yyyy format)
 - Status filter (All, Confirmed, Pending, Cancelled)
 - Search bar to find specific bookings
4. **Action:** Click the "+ New Booking" button to set up a new booking.
5. **Action:** Click "View" to see detailed booking information.
6. **Action:** Click "Export Booking PDF" to generate a report.
7. **Action:** Click "Reset" to remove all filters.
8. **System Response:**
 - Filters: Modifies the table according to chosen criteria
 - New Booking: Launches the booking creation form
 - View: Shows all booking details along with status information

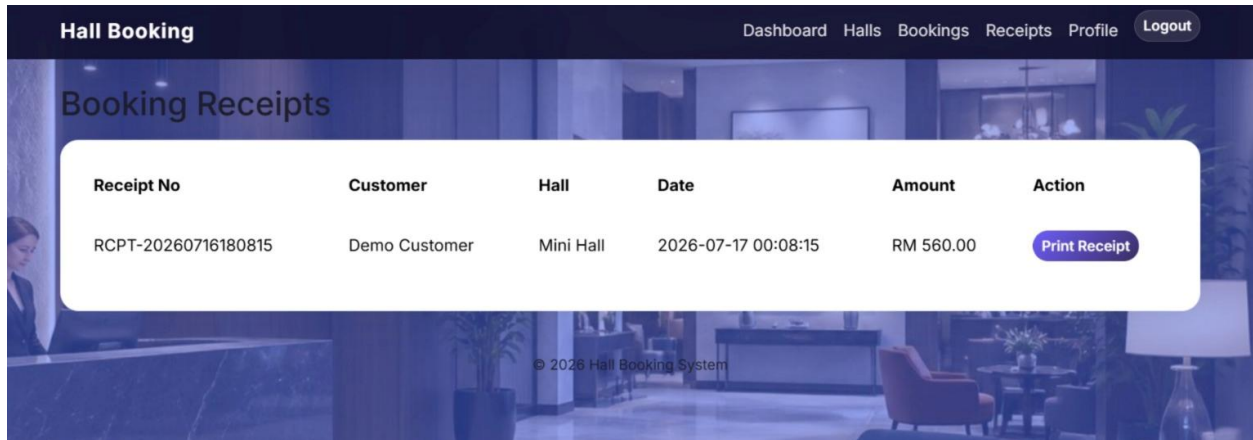


Figure 26: Booking Receipts (Customer View)

Booking Receipts Flow:

1. **Action:** The customer selects "Receipts" from the sidebar menu.
2. **System Response:** The customer's personal Booking Receipts page appears.

Receipt No	Customer	Hall	Date	Amount	Action
RCPT-2026071161808150815	Demo Customer	Mini Hall	2026-07-17 00:08:15	RM 560.00	Print Receipt

Table 8: Receipts Table

3. **Action:** The customer can select "Print Receipt" to create a PDF receipt.
4. **System Response:** The system creates a formatted receipt with payment and booking details.

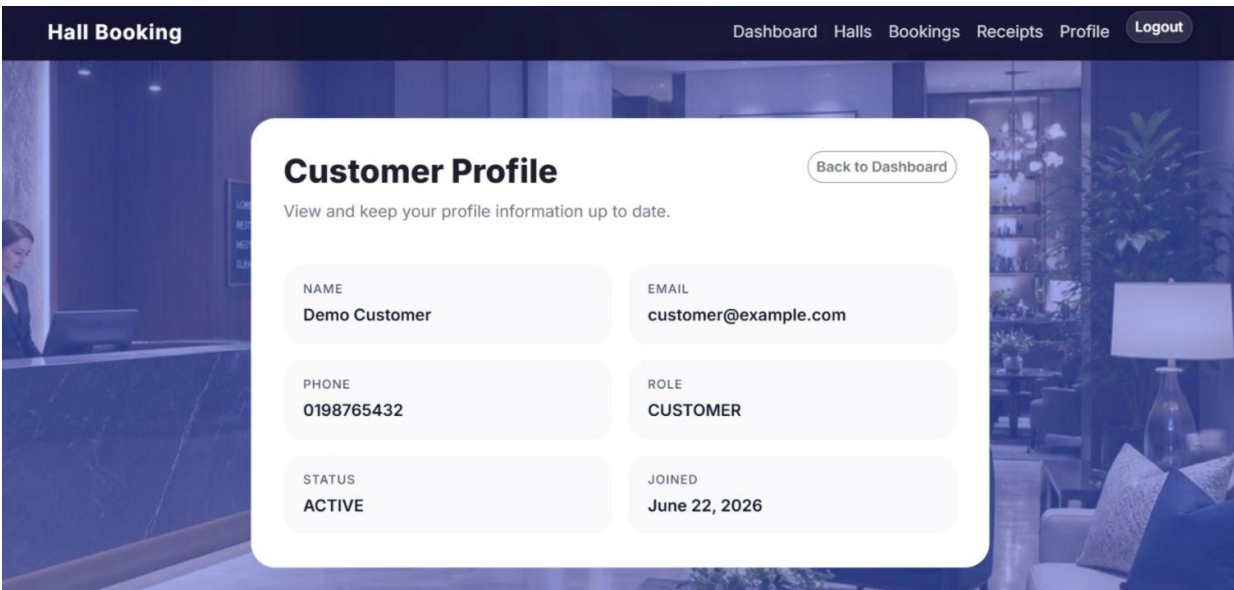


Figure 27: Customer Profile

Customer Profile Management Flow:

1. **Action:** Customer clicks on "Profile" from the sidebar menu.
2. **System Response:** The system shows the customer profile page.
3. **Action:** Customer can edit profile information (Name, Email, Phone).
4. **Action:** Click "New Booking" to make a reservation straight from the profile.
5. **Action:** Click "My Bookings" to view booking history.
6. **System Response:**
 - Updates saved to users' table
 - Redirects to booking creation or view pages accordingly

PDF Download Flow:

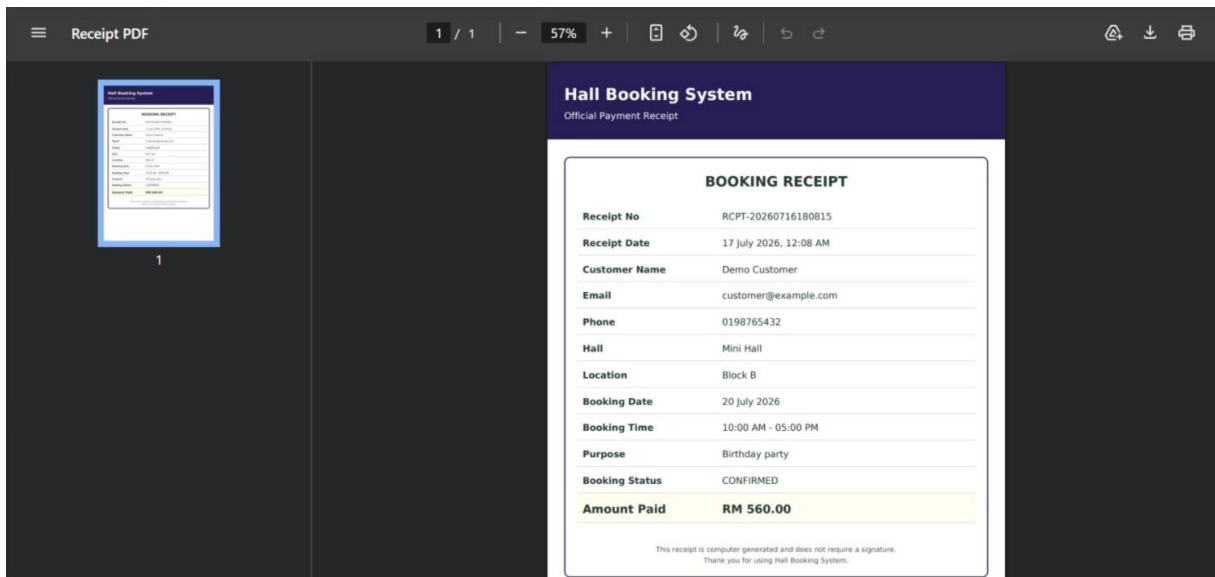


Figure 28: Hall Booking System receipt PDF

Step 1: Accessing the receipt

Customer Access:

1. The customer accesses the Hall Booking System.
2. Selects "Receipts" from the menu in the sidebar
3. A list of every booking receipt is shown by the system.

Admin Access:

1. The admin accesses the system
2. Selects "Receipts" from the menu in the sidebar
3. The system shows every customer's receipt.

Step 2: Locating the exact receipt

The information that displayed by the receipt:

- Receipt No: Special receipt identifier
- Customer: The customer's name who made the booking
- Hall: Hall that was booked
- Date: The time and date the receipt was created

- Amount: Total amount that paid for the booking
- Action: Print Receipt button

Step 3: Initiating PDF Download

1. **Action:** The user selects the preferred receipt by clicking the "Print Receipt" button next to it.
2. **System Reaction:**
 The system retrieves the receipt of information from the database.
 Produces a PDF file containing full receipt information.
 Download the PDF directly or open it in a new browser tab.

3.4 Customer Key Features Summary

Feature	Description
Registration	Create new customer account and assign a role
Login	Login using secure authentication to access the system
Browse Halls	See all the available halls together with their location, rate and capacity information
Search Halls	Sort halls by location or name
Create Bookings	Make a reservation, choose the hall, time and date then send in the request
View Bookings	Keep track of all individuals' reservations with progress updates
Filter Bookings	Bookings can be filtered by search terms, date, or status
View Receipts	View, access, and print receipts for reservations
Profile Management	Update personal data
Real Time Status	View the booking status in real time (Pending, Confirmed, Cancelled)

Table 9: Key Features Summary for Customer

Customer Booking Flow Summary:

- **Registration:** The customer registers an account
- **Login:** The customer enters the system
- **Browse Halls:** The customer views the halls that are available to book
- **Select Hall:** The customer clicks “Book” after selecting a hall
- **Enter Booking Details:** The customer chooses the purpose, date, and time
- **Submit Booking:** The system generates a “Pending” booking
- **Track Status:** The customer uses the dashboard to keep track of the booking status
- **Get Confirmation:** The status is changed to “Confirmed” if admin authorises the reservation
- **View Receipt:** The customer can view and print the reservation receipt

4.0 Identify Data

The data involved in the system are:

- User information
- Customer details
- Admin details
- Hall information
- Booking records

5.0 Data Dictionary

The **users** table stores information about all users who access the Hall Booking System. This includes the system administrator and customers. The table helps the system identify user roles, manage login details, store contact information, and control whether a user account is active or inactive.

5.1 Table: users

Attribute	Data Type	Description	Key
user_id	Int Auto_Increment	Unique ID for each user	Primary key
name	Varchar (100)	Full name of the user	Not null
email	Varchar (100)	User email address for login and contact	Unique, not null
password	Varchar (255)	User password stored in encrypted form	Not null
role	Enum('Admin','Customer')	Type of user such as admin or customer	Not null, default: CUSTOMER
phone	Varchar (20)	User phone number	-
approval_status	Enum('Active','Inactive')	Approval conditions such as pending, approved, or rejected.	Not null, default: ACTIVE
created_at	Timestamp	Date and time when the user	Default: CURRENT_TIMESTAMP

		account created	was	
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Table 10: Users

The halls table stores details about the halls available for booking in the system. It includes information such as hall name, location, capacity, description, hourly rental rate, and hall status. This table helps customers view available halls and helps the administrator manage hall information.

5.2 Table: halls

Attribute	Data Type	Description	Key
hall_id	Int Auto_Increment	Unique ID for each hall	Primary key
location	Varchar (150)	Location of the hall	Not null
capacity	Int	Maximum number of people allowed in the hall	Not null
description	Text	Detail or information about the hall	-
hourly_rate	Decimal (10,2)	Rental price of the hall per hour	Not null
status	Enum('Active','Inactive')	Hall availability status such as available, unavailable, or maintenance	Not null
create_at	Timestamp	date and time when the hall record was created	Not null
updated_at	Timestamp	date and time when the hall information was last updated	Not null

Table 11: Halls

5.3 Table: booking_receipts

Attribute	Data Type	Description	Key
receipt_id	INT Auto_Increment	Unique ID for each booking	Primary Key
booking_id	INT	ID of the booking which this receipt is linked to	Foreign Key
receipt_date	DATETIME	Date and time when the receipt was generated	Not Null
amount_paid	DECIMAL (10,2)	Total amount paid for every booking	Not Null
receipt_number	VARCHAR (50)	Special reference number for every receipt	Not Null

Table 12: Booking Receipts

5.4 Table: bookings

Attribute	Data Type	Description	Key
booking_id	INT Auto_Increment	Unique ID for each booking	Primary Key
user_id	INT	ID of the customer who made the booking	Foreign Key
hall_id	INT	ID of the hall being booked	Foreign Key
booking_date	DATE	Date of the booking event	Not Null
start_time	TIME	Start time of the booking	Not Null
end_time	TIME	End time of the booking	Not Null
purpose	VARCHAR (255)	Purpose of the booking (e.g., 'Seminar', 'Wedding')	-
booking_status	ENUM ('Pending', 'Confirmed',	Current status of the booking request	Not Null, default: 'Pending'

	'Rejected', 'Cancelled')		
created_at	TIMESTAMP	Date and time the booking record was created	Default: CURRENT_TIMESTAMP
updated_at	TIMESTAMP	Date and time the booking record was last updated	-

Table 13: Bookings

6.0 Keys

Primary Key

Primary Key is a unique identifier used to identify each record in a table.

Examples:

- user_id
- hall_id
- receipt_id
- booking_id

Foreign Key

Foreign Key is an attribute used to connect one table with another table.

Examples:

- user_id in table bookings
- hall_id in table bookings

7.0 System Flow

1. The customer and admin register and log into the system using a username and password.
2. The customer browses available venues and selects a suitable venue hall.
3. The customer enters hall booking details such as date, time, and purpose.
4. The system stores the booking information on the booking table.
5. The admin reviews and manages the booking request.
6. The admin verifies and processes the booking information.
7. After booking verification, the booking status is updated to be confirmed.
8. The admin can monitor all bookings, and venue availability through the system dashboard.

7.1 Flowchart table

Hall Booking System Flowchart

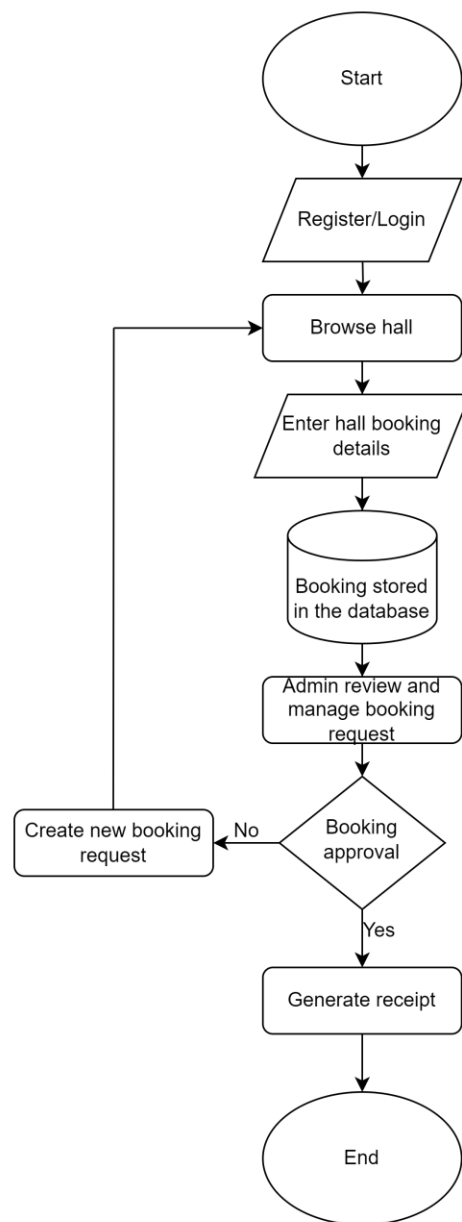


Figure 29: Hall Booking System Flowchart

7.2 Hall Booking System Entity Relationship Diagram (ERD)

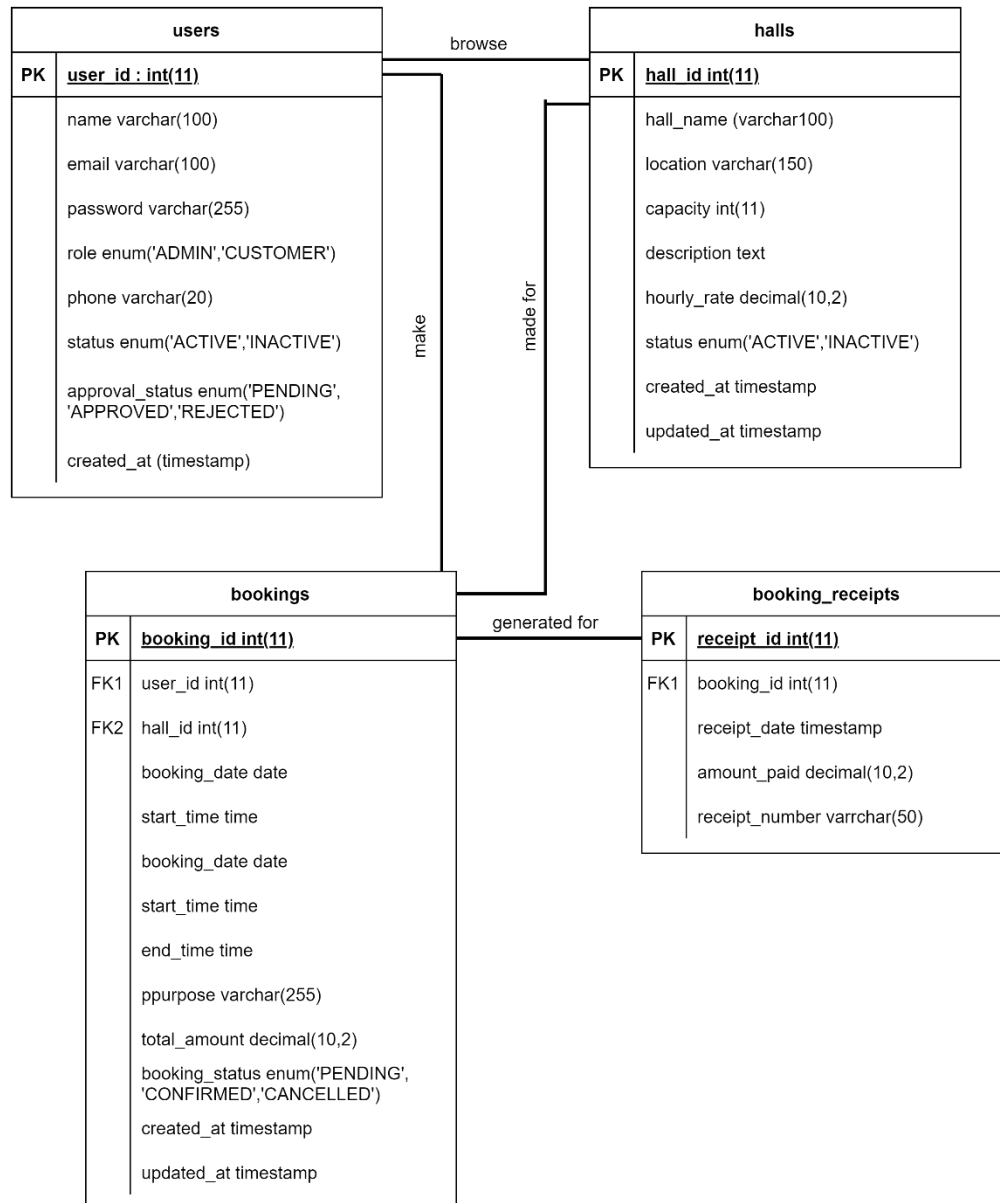


Figure 30: Hall Booking System ERD

7.2.1 Entity Relationships:

1. One customer can make many bookings (1:M)
2. One customer can browse many halls (1:M)
3. One booking made by one customer (1:1)
4. One receipt generated for one booking (1:1)

8.0 Team Roles and Contributions

The Hall Booking System was successfully developed due to the effort put forth by four members in the team. Each of the members was allocated specific roles in the planning, development, testing, and documentation stages of the project. While each of the members worked on distinct roles, all the members cooperated in order to ensure successful completion of the project.

Table 1: Summary of Individual Roles and Contributions

Nur Irdiana Binti Moktar Project Manager • Coordinated all aspects of the project. • Monitored the progress of the project and tasks. • Managed documentation and made sure deadlines were met. • Conducted final review of the system before submitting.	Siti Zulaikha Binti Mohd Salleh System Developer • Developed the Hall Booking System using PHP, HTML, CSS, Bootstrap, and MySQL. • Created the CRUD operations and booking capabilities. • Created the receipt generator in PDF format. • Carried out testing of the system and fixed system bugs. • Helped in preparing the technical documentation.
Intan Nursakinah Binti Idrus UI/UX Designer • Designed the interface of the system. • Developed responsive and user-friendly designs. • Enhanced user experience. • Helped in interface testing and maintaining consistency in interface.	Nurul Hidayah Binti Anuar System Analyst • Analyze the user requirements and system requirements. • Conducted analysis documentation for the system. • Designed the database and data dictionary. • Aided in conducting ERD documentation.

9.0 Contact Information

Should any inquiries, technical support, feedback, or information about the Hall Booking System be required, the members of the project team can be reached using the following contact details.



Name: Nur Irdiana binti Moktar
Email: 2024244524@student.uitm.edu.my



Name: Siti Zulaikha binti Mohd Salleh
Email: 2024680156@student.uitm.edu.my



Name: Intan Nursakinah binti Idrus
Email: 2024220646@student.uitm.edu.my



Name: Nuri
Email: 2024680246@student.uitm.edu.my

10.0 Conclusion

The Hall Booking System is suggested as a reliable way to update and simplify the hall reservation procedure. The system aims to greatly improve the experience for admins and users by switching from standard manual techniques, which are frequently challenged by errors and inefficiencies, to a centralized web-based platform. Managing user data, hall information, and booking records is made possible by the system's design, which is described in the data dictionary and ERD. This will enable users to check real-time availability and make reservations simply while admin can efficiently manage hall details, verify bookings, and generate receipts. This digital approach not only reduces the risk of human error, such as bookings redundancy, but also improves communication, reporting, and overall operational efficiency, thereby fulfilling the primary objectives.

10.1 Limitations

The Hall Booking System's current design has a number of limitations that should be noted despite its potential. These restrictions offer opportunities for further advancement:

- **Limited Payment Integration:** There isn't an integrated online payment gateway in the system as described. Payments may still need to be made offline because the booking_receipts database has an amount_paid field but does not outline a secure transaction procedure.
- **Lack of Real-Time Notification System:** The system lacks a real-time notification system (such as SMS or email alerts). After an administrator's permission, customers would have to manually monitor the system for updates on their booking status, which is not ideal.
- **Management of Users Scope:** Viewing and deactivating accounts may be the extent of the admin user management abilities. Role-based access, more precise permissions, and user activity logging are common features of full-featured user management systems which are not covered here.

- **Support for Mobile Platforms:** The system is characterized as a web-based application, which could not be completely mobile-friendly. The user experience on smartphones and tablets may be limited by the lack of a mobile application or a fully responsive design.

10.2 Future Improvements

A number of future enhancements are suggested in order to overcome the existing constraints and increase the system's functionality. These enhancements would bring the system into compliance with both user expectations and current technology trends:

- **Integration of Online Payment Gateway:** The booking process should include a secure payment system (such as linking with PayPal, Stripe, or a local bank API). This will speed up the process and increase the accuracy of financial records by allowing clients to make payments immediately upon booking confirmation.
- **Real-Time Notification Implementation:** A notification engine that automatically notifies clients via email or SMS should be added to the system. To improve user experience, an email can be sent, for example, when a reservation is successfully made, accepted, or cancelled.
- **Improved User and Security Management:** To improve system security and accountability, subsequent updates should incorporate more advanced user management capabilities like role-based access restriction, multi-factor authentication (MFA), and a complete audit history of admin operations.
- **Development of a Mobile Application:** Making sure the online application is properly responsive or developing a companion mobile application would greatly increase accessibility, enabling clients to easily browse, book, and manage their bookings from any device.

- **Advanced Analytics and Reporting:** To enable data-driven decision-making, a more comprehensive reporting module might be created to give admins insightful information including booking trends, peak usage periods, hall utilization rates, and income reports.

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